

PROJECT SPECIFICATION DOCUMENT FOR MSCI 436 - DECISION SUPPORT SYSTEMS

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1 Team-based Project Specifics

Each team will work towards *operationalizing* (i.e., deploying into a functional, interactive system) a data-driven Intelligent Decision Support System (IDSS) for an engineering or a business decision making task. This is not a modeling project, and teams are free to choose any data sources and any models which are appropriate to the IDSS task. Justifying the use of the model, and its alignment with the decision making task is key. The model may be based on ML, OR, or any other disciplines which you may have learned in any courses or something that you found via your own research. You will be working on applying this model on real-world data to automate a process and draw insights for stakeholders. Teams are encouraged to talk to subject matter experts, faculty members to help shape their projects.

Tutorials for formative feedback and bonus for subject matter expert involvement: Teams are encouraged to talk to the teaching staff regarding any aspects of their projects to gain formative (ungraded) feedback. Teams that demonstrate engagement with a subject matter expert or faculty member beyond the teaching staff — naming the expert and describing how the consultation shaped the use-case, data choice, or system design — will receive one bonus mark on the proposal.

The IDSS Requirement. Your IDSS must place a human in the loop: the user should be able to change inputs, parameters, or scenarios that feed into the model and see how the decision-relevant outputs respond. This interactivity must be load-bearing — it should change what decision the stakeholder would make, not merely change how data is displayed. Before submitting, ask yourself: could a scheduled script and an emailed report do what our system does? If the answer is yes, you have built an automated pipeline, not a decision support system.

1.1 Team Formation Deadline

- Students will form 4 person teams and post their team composition to Campuswire by the team formation deadline.
- Due to the class strength some teams will be allowed to have 5 members, this will be first-come-first-serve and at the discretion of the teaching staff.
- Final teams assignments along with a Team Number (randomly assigned) will be announced on Campuswire; It is team/student's responsibility to get back to us with any discrepancy regarding team memberships within two days of this announcement.

1.2 Proposal

Teams will submit the design worksheet for IDSS design as part of the proposal (see attached). This is a worksheet in which the information needs to be filled out. If a team requires additional space, one extra sheet (A4 Times New Roman/Arial, font size 12) is permitted. Teams will identify a value proposition identifying a particular way in which their DSS will help a stakeholder. Note that the use-case you identify is the most important part of the design, and the way you leverage the data and model you have identified to help the stakeholder will be a major component in grading.

1.3 Slides-style Report

The teams will submit a slides-style report (pdf, 12 slides max with references). See the requirement regarding an additional slide on AI disclosures in the next section (does not count towards the slides limit).

- What problem does your IDSS solve?
- Why does your problem necessitate use of a IDSS?
- Data – Describe the data and the processing involved. Why is the data appropriate for the identified task? What are the limitations?
- Model – Justify the use of the data-driven model for the task. What are the advantages and disadvantages of this model?
- User-Interface demo – What control does the user have to analyze the results of the model? How will it help them to make decisions identified in your proposal? Visualizations must be tied to the specific decision identified in your proposal — general data displays do not meet this requirement.
- Operationalization – What is your operationalization and deployment plan? What kind of infrastructural and software needs of the IDSS?
- Github - You also need to include a github link to your codebase.

Quality of slides, problem identification, need for IDSS, justification of data and model, quality of user-interface and visualizations therein will be used for evaluation.

1.4 In-class Presentation

Teams will present and demo their IDSS in the class. Teams are highly encouraged to use the slides developed as part of the report, and each team member is required to present. The presentation should be at most 10 minutes long, plus 1-2 minutes Q&A. The 10 minute duration is a hard stop to keep the presentations within the class time, and teams will be penalized for going over time. *Note that a team may be asked to present in the next lecture slot if for any reason we run over class time.* This means that you are expected to be ready to present in the next class if required.

Teams are strongly advised to prepare a contingency plan in case of technical failure (e.g. a video recording) during the live demo. A system that fails to run during the presentation will receive zero marks on the Demo criterion regardless of the reason.

2 Artificial Intelligence Tools

This course is about building systems that support human decision-making. That framing applies to how you use AI on this project: *you are the decision-maker, the AI is a tool you direct and verify.* Thoughtful, disclosed AI use is a professional skill worth developing. Undirected or undisclosed use undermines the learning the project is designed to produce.

Permitted uses include code implementation and debugging, writing polish, visualization generation, and literature search.

What must remain the team's own reasoning. The use-case and stakeholder decision, the argument for why an IDSS is required, the model justification, and the operationalization plan are the intellectual core of the project and its primary grading basis. AI may inform your thinking on these points, but the reasoning and judgement must be yours.

AI Disclosure slide (report) and Code development Include one slide titled *AI Use Disclosure* — it does not count toward the 12-slide limit. List each tool used, the task it assisted with, and in one sentence what the team changed, added, or verified. If no AI tools were used, the slide must say so explicitly. Omitting this slide will result in a penalty of 2 pts from the P3 raw score. Your repository should show commits distributed across the project period; a single large commit before the deadline is inconsistent with genuine development.

Your responsibility. You are accountable for the accuracy and quality of all submitted content regardless of how it was produced. Undisclosed AI use discovered through inconsistency between written work and demonstrated understanding (e.g. Q&A) will be referred to the academic integrity process; point deductions do not apply in its place.

3 Rubrics

Grading Rubric

The project is graded across three deliverables. For every deliverable the course contribution equals the raw score divided by 2. The combined project weight is 30 % of the course grade.

Deliverable	Raw total	Course weight
P2 : Proposal	/30	15 %
P3 : Slides Report	/20	10 %
P4 : In-class Presentation	/10	5 %
Total		30 %

P2 : Proposal

/30 ÷ 2 = 15 %

Your proposal is the ML Canvas design worksheet. The use-case you identify is the most important part: the specific decision your IDSS will support and the stakeholder who will use it. The way you connect your chosen data and model to that decision will be heavily weighted.

1. Value proposition & decisions

/10 pts

Canvas sections: Value Proposition, Decisions

Level	What we are looking for
Excellent [10 pts]	A named stakeholder faces a specific, real decision. The Decisions box explains the exact mechanism by which the model's output changes what the stakeholder does, including concrete operational parameters (e.g. thresholds, trigger conditions, frequencies). The value proposition connects the system's output to a measurable benefit for the stakeholder — not just “better decisions” but a quantifiable or clearly bounded improvement.
Proficient [7 pts]	The stakeholder and the decision they face are identified. The Decisions box describes how the system helps at a functional level, but the operational parameters (how the output translates into an action) are missing or vague. The benefit stated in the value proposition is plausible but qualitative rather than measurable.

Developing [4 pts]	The value proposition describes a broad domain problem (e.g., “reduce supply chain costs”) rather than a specific decision. A stakeholder may be named but their workflow is not connected to the system’s outputs. The Decisions box is generic, restates the prompt, or could apply to any OR/ML system in the domain.
Weak [0 pts]	No stakeholder or decision is identified. The system is described as a model or analytical tool rather than a decision support system. The value proposition is indistinguishable from a generic data science project description.

2. Data**/6 pts***Canvas sections: Data Sources, Data Collection*

Level	What we are looking for
Excellent [6 pts]	Data sources are named with a specific access method (e.g., API endpoint, database table name, website to scrape, file format). The collection strategy addresses both the initial training set <i>and</i> how data will be updated as the system operates — the evolving, non-static nature of the data is made explicit. At least one practical constraint (rate limits, labelling cost, privacy restriction, collection frequency) is acknowledged.
Proficient [4 pts]	Sources are identified and the access method is plausible. The collection plan addresses either the initial training set <i>or</i> ongoing updates, but not both. Constraints are acknowledged but not elaborated.
Developing [2 pts]	Data sources are named but described vaguely (e.g., “a public dataset,” “Kaggle,” “company data”). There is no collection strategy and no acknowledgement of how the data will be updated over time.
Weak [0 pts]	No data sources are identified, or the data described is clearly unavailable, fabricated, or inappropriate for the stated task.

3. Prediction & model**/6 pts***Canvas sections: Prediction Task, Building Models, Features*

Level	What we are looking for
Excellent [6 pts]	The prediction task is fully specified: task type (classification/regression/etc.), the entity on which predictions are made, the possible outcomes, and the wait time before outcomes can be observed. The model approach is named and plausible for the task. Features are described specifically as the inputs that will be <i>available at prediction time</i> — not just what exists in the raw data.
Proficient [4 pts]	The task type and model approach are identified and the pairing is sensible. One element — feature specification, wait time, or outcome definition — is incomplete or left vague.

Developing [2 pts]	A model type is mentioned but the task specification is vague (e.g., “predict customer behaviour”). Features are described generically (“relevant variables,” “input data”) with no reference to what will actually be available at prediction time.
Weak [0 pts]	No task specification or model is identified, or the model choice is clearly misaligned with the stated problem.

4. User interface

/5 pts

Canvas section: User Interface

Level	What we are looking for
Excellent [5 pts]	All four sub-questions are answered with task-specific content: (1) the specific insights the user needs to make the decision, (2) the metric types and visualization approaches that will present those insights, (3) which hyperparameters or model settings the user can adjust for their analysis, and (4) what inputs the user can provide to the system. Answers go beyond generic descriptions and are grounded in the identified use-case.
Proficient [3 pts]	Three of the four sub-questions are answered with specificity. The missing one is absent rather than vaguely attempted.
Developing [1 pt]	The UI is described at a concept level (e.g., “a dashboard with charts and filters”) but fewer than three sub-questions are specifically answered. Answers are generic rather than tied to the identified decision and stakeholder.
Weak [0 pts]	The User Interface section is blank, contains only the printed prompt, or describes a static report rather than an interactive interface.

5. System completeness

/3 pts

Canvas sections: Impact Simulation, Making Predictions, Monitoring — 1 point per section with substantive content

Level	What we are looking for
Excellent [3 pts]	All three sections contain substantive, task-specific content. Each section answers at least one of its prompt questions with specificity that goes beyond reprinting the question as the answer.
Proficient [2 pts]	Two of the three sections have substantive content. One section contains only a generic or one-line response.
Developing [1 pt]	One section has substantive content. Two sections are blank or contain only the printed prompt.
Weak [0 pts]	All three sections are blank or contain only the printed prompts with no added content.

Bonus (+1 pt): Teams that demonstrate engagement with a subject matter expert or faculty member *beyond* the teaching staff receive one bonus point on the proposal. To qualify, the expert

must be named and you must describe how the consultation shaped your use-case, data choice, or system design. A general mention of “consulting an expert” does not qualify.

Extra sheet: One additional A4 page in Times New Roman or Arial, 12 pt, is permitted. Content submitted outside this format will be disregarded.

P3 : Slides Report

/20 ÷ 2 = 10 %

Submit a PDF slide deck of at most 12 slides (excluding a reference slide). Include a working link to your GitHub repository. Your slides will be evaluated on: problem identification, the necessity of an IDSS, justification of data and model, quality of the user interface and visualizations, and your operationalization plan.

1. What problem does your IDSS solve?

/3 pts

Level	What we are looking for
Excellent [3 pts]	A named stakeholder, the specific decision they face, and the consequences of making a wrong decision are all stated clearly. The scope is focused: neither so trivial that an IDSS is unnecessary nor so broad that the team could not realistically build it.
Proficient [2 pts]	The stakeholder and the decision are identified. The consequences of a wrong decision are left implicit, or the stakeholder is described somewhat generically (e.g., “a supply chain manager at a retail firm”) while the decision itself is specific.
Developing [1 pt]	The problem is described at a domain level (e.g., “our system optimizes supply chain efficiency”) but neither a specific stakeholder nor their particular decision is articulated.
Weak [0 pts]	The problem is absent, is purely a modelling exercise with no decision-making framing, or is so vague it could describe any engineering project.

2. Why does your problem necessitate an IDSS?

/3 pts

Level	What we are looking for
Excellent [3 pts]	You argue convincingly that interactive, ongoing decision support is required — not a one-time model run or a static report. Both the need for <i>user control over outputs</i> (the stakeholder must be able to adjust parameters and scenarios) <i>and</i> the use of <i>evolving, non-static data</i> are explicitly addressed and integral to the argument. A reader should be unable to substitute a scheduled batch pipeline or static dashboard for your system and still satisfy the requirement.
Proficient [2 pts]	The justification is sound and references either interactivity or evolving data, but addresses only one of the two. Your system is distinctly an IDSS — it could not plausibly be replaced by a static pipeline — but the complete argument for why is not made.
Developing [1 pt]	An IDSS is stated as the chosen approach but the justification is generic (e.g., “AI can help managers make better decisions”). The same argument would justify a static model, an automated report, or a scheduled batch process.

Weak [0 pts]	No justification is provided. The system is described as a model or a static reporting tool. The question of <i>why</i> an IDSS is not engaged with.
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3. Data**/4 pts**

Level	What we are looking for
Excellent [4 pts]	The data source, structure, volume, and evolving/non-static nature are all described. Preprocessing steps are explained. Limitations — including potential sources of bias, data gaps, and quality issues — are honestly assessed rather than minimized or omitted.
Proficient [3 pts]	Most elements are covered. One aspect — typically the evolving-data justification or the limitations — receives only surface treatment (e.g., “the data may have some missing values” without elaboration).
Developing [2 pts]	The data is named and a source is given but the description stays at a high level. Volume, structure, or preprocessing are mentioned but not explained. Limitations are absent.
Weak [0 pts]	The data is not described, is not real data, or is clearly inappropriate for the identified task.

4. Model**/4 pts**

Level	What we are looking for
Excellent [4 pts]	The model choice is justified against your specific task — not merely the general domain. Trade-offs are evaluated from both sides (advantages and disadvantages). How the model handles evolving or non-static data is directly addressed. Note: this project is <i>not</i> a modelling exercise — you are assessed on justification and fit, not on technical sophistication.
Proficient [3 pts]	The justification is task-linked and reasonable. Trade-offs are noted but one side (usually the advantages) is more thoroughly treated than the other.
Developing [2 pts]	The model is named and briefly described but the justification is domain-level (e.g., “neural networks are good at classification”) rather than specific to your task. Trade-offs are one-sided or generic.
Weak [0 pts]	No justification is provided. The model is treated as a black box, or the model choice is clearly misaligned with the decision task.

5. User interface & code**/3 pts**

Level	What we are looking for
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Excellent [3 pts]	The user interface provides <i>load-bearing</i> controls: when the user changes an input or adjusts a parameter, the model output changes in a way that affects the decision. Visualizations are explicitly tied to the decision identified in your proposal — the audience can see how a change in output would alter the stakeholder’s choice, not merely observe data. Your GitHub link is working and the code is organized, documented, and runnable by a third party without oral instruction.
Proficient [2 pts]	The interface is functional and controls are present. Visualizations are relevant but the connection to the proposal decision is implicit — the audience must infer the decision implication rather than read it directly from the visualization. GitHub is accessible but partially disorganized or missing a README.
Developing [1 pt]	Controls are <i>cosmetic</i> : they reformat or filter data displays without affecting model outputs or decision-relevant results. Visualizations show data but are not connected to the decision identified in your proposal. GitHub is present but disorganized or partially missing.
Weak [0 pts]	No interactive interface is present. The GitHub link is missing, broken, or the repository is empty.

6. Operationalization

/3 pts

Level	What we are looking for
Excellent [3 pts]	All four elements are addressed: (1) how the data pipeline continues to function after deployment (not just at launch), (2) when and how the model will be updated or retrained, (3) how users will access the system, and (4) the hardware and software infrastructure the system requires.
Proficient [2 pts]	Three of the four elements are present. Either the infrastructure requirements or the ongoing maintenance aspect (pipeline continuity or retrain schedule) is underdeveloped.
Developing [1 pt]	Deployment is mentioned with a named platform (e.g., “we would host this on AWS”) but there is no detail on data continuity, update frequency, or infrastructure configuration.
Weak [0 pts]	No operationalization section is present, or the section contains a single vague sentence with no substance.

Slide quality deduction (−1 pt): A deduction of one point is applied after scoring if any of the following apply: the deck exceeds 12 slides (excluding references), figures are consistently unlabelled, or slides are predominantly blocks of text with no meaningful visual content and are hard to read.

P4: In-class Presentation and Demo

/10 ÷ 2 = 5 %

Your team will present and demo your IDSS live in class. The presentation should be at most 10 minutes, followed by 1–2 minutes of Q&A. The 10-minute limit is a hard stop — teams will be penalized for going over time. You are expected to be ready to present in the next lecture slot if required. It is recommended that you use the slides submitted as your report; minor formatting updates between submission and presentation are permitted.

1. Live demo**/4 pts**

Level	What we are looking for
Excellent [4 pts]	The system runs live during the presentation. At least two distinct user interactions are demonstrated — for example, changing an input value, adjusting a model parameter, and exploring a different scenario. Each interaction's output is interpreted in decision terms: the presenter explicitly connects the result to the choice the stakeholder would make <i>differently</i> as a consequence. Showing “here is the chart” without explaining its decision implication does not qualify.
Proficient [3 pts]	The demo runs live and at least one user interaction is demonstrated. The team provides at least a partial interpretation of what the output means for the stakeholder's decision — the audience is not left to infer all decision implications entirely on their own.
Developing [2 pts]	The demo runs live but outputs are presented without any interpretation; the audience must infer all decision implications unaided. <i>OR</i> : the demo is pre-recorded, consists of screenshots, or is a fixed walkthrough with no live interaction.
Weak [0 pts]	No demo is provided. The system does not run. Only slides, mockups, or a verbal description of what the system would do are presented. Teams are strongly advised to prepare a contingency plan: a system that fails to run during the presentation receives zero marks on this criterion regardless of the reason.

2. Narrative & communication**/3 pts**

Level	What we are looking for
Excellent [3 pts]	The presentation opens by grounding the audience in the specific stakes and decision — not just the topic area. Problem, data, model, and demo are connected by a single logical thread where each section builds on the previous. Transitions between speakers are seamless and clearly prepared.
Proficient [2 pts]	The opening establishes the problem and the narrative is mostly coherent, but one section feels disconnected from the rest or one speaker transition is abrupt or visibly unprepared.
Developing [1 pt]	The required content is covered but without a narrative thread. The presentation feels like the report sections being read aloud in sequence. Transitions between speakers are abrupt throughout or preparation is clearly uneven across members.
Weak [0 pts]	There is no coherent structure. The audience cannot follow the relationship between the problem, the system, and the decision. The presentation is clearly unrehearsed.

3. Technical defence & Q&A**/3 pts**

Level	What we are looking for
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Excellent [3 pts]	Data and model choices are defended with specificity — the team articulates <i>why this data and this model for this task</i> , not just what they are. Limitations are raised voluntarily without prompting. Questions from the audience are answered accurately and with depth; each team member is able to field questions about their own contribution.
Proficient [2 pts]	Justification is present and task-linked. Either the data or the model is defended more thoroughly than the other. Limitations are acknowledged only after prompting. Most questions are answered correctly; one answer is incomplete or one team member is unable to contribute meaningfully.
Developing [1 pt]	Data and model are described (inputs, outputs, source) but not genuinely defended. Limitations are absent from the presentation. Some questions are answered; at least one answer is factually incorrect or deflected; technical questions are deferred to a single team member.
Weak [0 pts]	No verbal justification is provided for the data or model choices. The team cannot answer questions accurately; responses are inaccurate, missing, or entirely deferred to one person.

Time deduction (applied after scoring):

Finish time	Deduction
$\leq 10:00$ (on time)	0 pts
10:01–10:30	−0.5 pts
10:31–11:00	−1 pt
11:01–11:30 (hard stop)	−1.5 pts